STAT 2020 Discussion
Week 5

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February 16, 2026

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Exam 1 Reminders

- Exam 1: **next Monday** (8:30–9:20am), in the lecture room
- Must submit by 9:20am (another class starts at 9:30am)
- Coverage: up to and including **Binomial distribution**
- Bring: calculator; (and one cheat sheet if allowed)

Bubble Sheet (Important)

- Fill out: **Name, ID, Section, Date, Version**
- Wrong version may result in a zero
- Demonstration: only first 15 questions need to be filled

PMF and CDF (Discrete)

- PMF: $f(x) = P(X = x)$
- CDF: $F(t) = P(X \leq t)$
- For discrete X, $F(t)$ is a **step function**

PMF → CDF (Recipe)

$$F(t) = P(X \leq t) = \sum_{x \leq t} f(x).$$

- Sort support values
- Running sum ⇒ piecewise CDF

Example 1: PMF (Flip a Coin Twice)

Let X = number of heads when flipping a fair coin twice.

x	0	1	2
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Goal: write $F(t) = P(X \leq t)$.

Solution: Example 1 (CDF)

$$F(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{4}, & 0 \leq t < 1, \\ \frac{3}{4}, & 1 \leq t < 2, \\ 1, & t \geq 2. \end{cases}$$

Solution: Example 1 (CDF)

$$F(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{4}, & 0 \le t < 1, \\ \frac{3}{4}, & 1 \le t < 2, \\ 1, & t \ge 2. \end{cases}$$

$$P(X \le 1.5) = F(1.5) = \frac{3}{4}, \qquad P(X > 2) = 1 - F(2) = 0.$$

CDF → PMF (Jump Size)

For discrete X:

$$P(X = x) = F(x) - \lim_{t \uparrow x} F(t).$$

- "jump size at x" = probability mass at x

Expectation

$$E(X) = \sum_{x \in S_X} x f(x).$$

$$E[h(X)] = \sum_{x \in S_X} h(x) f(x).$$

Variance (Two Forms)

$$\mathrm{Var}(X) = E[(X-\mu)^2] = \sum_{x \in S_X} (x-\mu)^2 f(x).$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Example 2: Mean & Variance from a PMF

$$P(X = 0) = 0.7, \quad P(X = 1) = 0.2, \quad P(X = 2) = 0.1.$$

Compute $E(X)$ and $\mathrm{Var}(X)$.

Solution: Example 2

$$E(X) = 0(0.7) + 1(0.2) + 2(0.1) = 0.4.$$

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$$E(X) = 0(0.7) + 1(0.2) + 2(0.1) = 0.4.$$

$$E(X^2) = 0^2(0.7) + 1^2(0.2) + 2^2(0.1) = 0.6.$$

Solution: Example 2

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$$E(X^2) = 0^2(0.7) + 1^2(0.2) + 2^2(0.1) = 0.6.$$

$$\mathrm{Var}(X) = 0.6 - (0.4)^2 = 0.44.$$

$$\boxed{E(X) = 0.4,\ \mathrm{Var}(X) = 0.44}$$

Discrete Uniform

If X is uniform on $\{x_1, \ldots, x_N\}$:

$$P(X = x_i) = \frac{1}{N}.$$

Example: $X \in \{1, \ldots, 10\}$.

Bernoulli

$$X \sim \text{Bernoulli}(p), \quad P(X=1)=p,\ P(X=0)=1-p.$$

$$E(X)=p, \qquad \text{Var}(X)=p(1-p).$$

Binomial

$$X \sim \mathrm{Bin}(n, p)$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n.$$

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Example 3: Rolling a Die

Roll a fair die $n = 10$ times. Let $X =$ number of 6's.

$$X \sim \text{Bin}\left(10, \frac{1}{6}\right).$$

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$$P(X = 2) = \binom{10}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^8.$$

Example 4: Defective Parts

A manufacturer has 2% defective parts. Select $n = 30$ parts. Let $X =$ number of defective parts.

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$$E(X) = np = 30(0.02) = 0.6.$$

Wrap-Up: Key Takeaways

- PMF $\to$ CDF:

$$F(t) = \sum_{x \le t} P(X = x)$$

- CDF $\to$ PMF (jump):

$$P(X = x) = F(x) - \lim_{t \uparrow x} F(t)$$

- Mean/Variance:

$$E(X) = \sum x f(x), \qquad \mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

- Binomial recognition: fixed n, fixed p, independent trials

In-Class Attendance Check (Quick)

A manufacturer produces parts where 2% are defective. Suppose 30 parts are randomly selected. Let $X =$ number of defective parts.

Compute $E(X)$.

Please submit: Full Name; Section Number; Your Answer

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A manufacturer produces parts where 2% are defective. Suppose 30 parts are randomly selected. Let $X =$ number of defective parts.

Compute $E(X)$.

Please submit: Full Name; Section Number; Your Answer

Answer:

$$X \sim \mathrm{Bin}(30, 0.02), \qquad E(X) = 0.6.$$